No. of Experiment: ……

Name of the Experiment: Determination the force constant (Spring constant) of a helical spring by plotting a graph between load and extension..

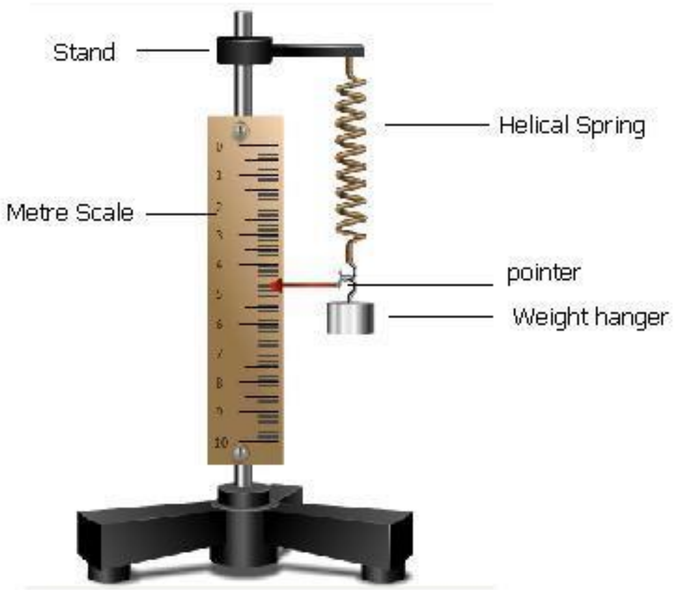
Theory:

What is a helical spring?

The helical spring, is the most commonly used mechanical spring in which a wire is wrapped in a coil that resembles a screw thread. It can be designed to carry, pull, or push loads. Twisted helical (torsion) springs are used in engine starters and hinges.

Let’s study how we can use the helical spring to do our experiment.

The helical spring is suspended vertically from a rigid support. The pointer is attached horizontally to the free end of



spring. A metre scale is kept vertically in such a way that the tip of the pointer is over the divisions of the scale; but does not touch the scale.

Helical spring works on the principle of Hooke’s Law. Hooke’s Law states that within the limit of elasticity, stress applied is directly proportional to the strain produced.

$$ie;\ Stress \propto Strain$$

$$\frac{Stress}{Strain} = a\ constant$$

When a load ‘F’ is attached to the free end of a spring, then the spring elongates through a distance ‘l’. Here ‘l’ is known as the extension produced. According to Hooke’s Law, extension is directly proportional to the load.

This can be represented as:

$$F \alpha l$$

$$F = kl$$

where ‘k’ is constant of proportionality. It is called the force constant or the spring constant of the spring.

A graph is drawn with load M in kg wt along X axis and extension, l in metre along the Y axis. The graph is a straight line whose slope will give the value of spring constant, k.

Materials Required:

- A spring
- A rigid support
- Weight hanger
- 50g or 20 g slotted weights
- A vertical wooden scale
- A fine pointer

Procedure:

1. The helical spring is suspended vertically from a rigid support. A pointer is attached horizontally at the free end of the spring.
2. A metre scale is kept vertically in such a way that the tip of the pointer is over the divisions of the scale, but does not touch the scale.
3. A dead weight, m_0 kg is suspended by the weight hanger to keep the spring vertical. The reading of the pointer on the metre scale is noted.
4. Now, gently add a suitable load of 50 g slotted weights to the hanger and the reading of the pointer is noted.
5. The weights are added one by one till the maximum load is reached. In each case, the reading of the pointer is noted.
6. The weights are then removed one by one and the reading of the pointer is noted in each case of unloading.
7. The average of the readings for each load during loading and unloading is calculated in each case. Let z0, z1, z2, z3…etc.., be the average readings of the pointer for the loads m_0, (m_0+50), (m_0+100), (m_0+150) etc.
8. From this, extension, l (in m) for the loads (m_0+50), (m_0+100), (m_0+150) etc. , are calculated as (z1-z0), (z2-z0), (z3-z0) respectively.
9. In each case, k =mg/l is calculated. The average value of k gives the spring constant in N/m.
10. A graph is drawn with load M in kg wt along X axis and extension, l in metre along the Y axis. The graph is a straight line. The reciprocal of the slope of the graph is determined. It gives spring constant in kg wt/m. The spring constant in N/m is obtained by multiplying this with g=9.8 m/s^2.

Observations:

Table for load and extension:

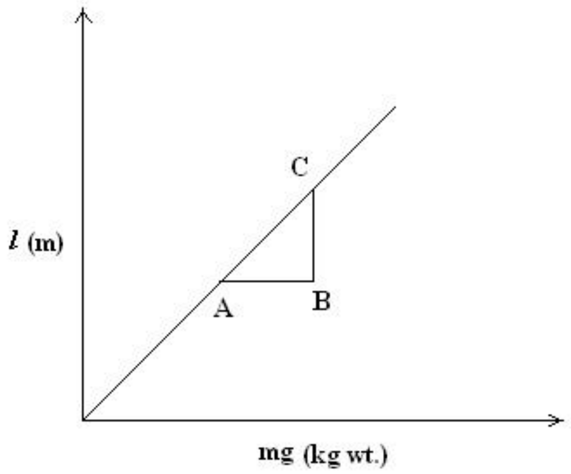
Serial No of Obs.	Load on hanger (m) (kg)	Tension, F= mg (N)	Reading of position of pointer tip			Extension l= z x 10^{-2} (m)	$k = \frac{mg}{l}$
			Loading X(cm)	Unloading Y(cm)	Mean, $z = \frac{x+y}{2}$ (cm)		
1	Dead load(m_0)						
2	(m_0+ .05)						
3	(m_0+.1)						
4	(m_0+.15)						
5	(m_0+.2)						

6	$(m_0+.25)$						
7	$(m_0+.3)$						

Mean k=…………N/m.

Calculations:

Spring constant, k from load extension graph



AB=---------kg wt

BC=---------m

$k = \frac{AB}{BC}$ = ---------Nm^{-1}

Result

By calculation, the force constant of the given spring = ...N/m.

From load-extension graph, the force constant of the given spring =……….N/m